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Calibration results
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Normalized Residuals
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Reprojection error (cam0):     mean 1.310613904935118, median 1.1328318000776956, std: 1.3677796345018147
Reprojection error (cam1):     mean 0.9829071507960704, median 0.8156919737713596, std: 0.8645296679904808
Gyroscope error (imu0):        mean 83.48266555883912, median 65.63841404974077, std: 61.55362787583873
Accelerometer error (imu0):    mean 1.3660989629244924, median 0.7573382920797183, std: 2.2192913031143053

Residuals
----------------------------
Reprojection error (cam0) [px]:     mean 1.310613904935118, median 1.1328318000776956, std: 1.3677796345018147
Reprojection error (cam1) [px]:     mean 0.9829071507960704, median 0.8156919737713596, std: 0.8645296679904808
Gyroscope error (imu0) [rad/s]:     mean 143.28419615850507, median 112.65748800998173, std: 105.64662773750497
Accelerometer error (imu0) [m/s^2]: mean 0.918419429987334, median 0.5091535982359727, std: 1.4920150800925152

Transformation (cam0):
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T_ci:  (imu0 to cam0):
[[ 0.01741959 -0.99984286 -0.00328745 -0.06266099]
 [ 0.0681459   0.00446755 -0.99766536  0.13712424]
 [ 0.99752328  0.0171549   0.06821301 -0.38712275]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam0 to imu0):
[[ 0.01741959  0.0681459   0.99752328  0.37791103]
 [-0.99984286  0.00446755  0.0171549  -0.05662271]
 [-0.00328745 -0.99766536  0.06821301  0.16300492]
 [ 0.          0.          0.          1.        ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.005897662180945326


Transformation (cam1):
```

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-----------------------
T_ci:  (imu0 to cam1):
[[ 0.03183194 -0.99947495 -0.00604586 -0.31740219]
 [ 0.07439261  0.00840137 -0.99719364  0.13469423]
 [ 0.99672086  0.03129284  0.07462098 -0.3797769 ]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[ 0.03183194  0.07439261  0.99672086  0.37861483]
 [-0.99947495  0.00840137  0.03129284 -0.30648285]
 [-0.00604586 -0.99719364  0.07462098  0.16073659]
 [ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
0.005739649382089708

Baselines:
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Baseline (cam0 to cam1):
[[ 0.99989227  0.00373575  0.01419481 -0.24976508]
 [-0.00382593  0.99997264  0.00633091 -0.00021516]
 [-0.01417077 -0.00638453  0.99987921  0.00728661]
 [ 0.          0.          0.          1.        ]]
baseline norm:  0.24987143477713597 [m]


Gravity vector in target coords: [m/s^2]
[-0.2190391  -9.7888283  -0.54706974]


Calibration configuration
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cam0
-----
```

Camera model: pinhole
Focal length: [1746.4539465334938, 1744.4669421726232]
Principal point: [958.9950621906956, 562.5906820013532]
Distortion model: radtan
Distortion coefficients: [-0.25701113919362983, 0.10262664411100493, 0.00043516222338876745,
-2.646838006691349e-05]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

cam1

Camera model: pinhole
Focal length: [1744.9386686543533, 1742.546552845488]
Principal point: [939.6720202553483, 569.8432138862356]
Distortion model: radtan
Distortion coefficients: [-0.25645546865320074, 0.1025772034872798, -0.00037405453354943558,
-0.00022666394863838954]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

IMU configuration
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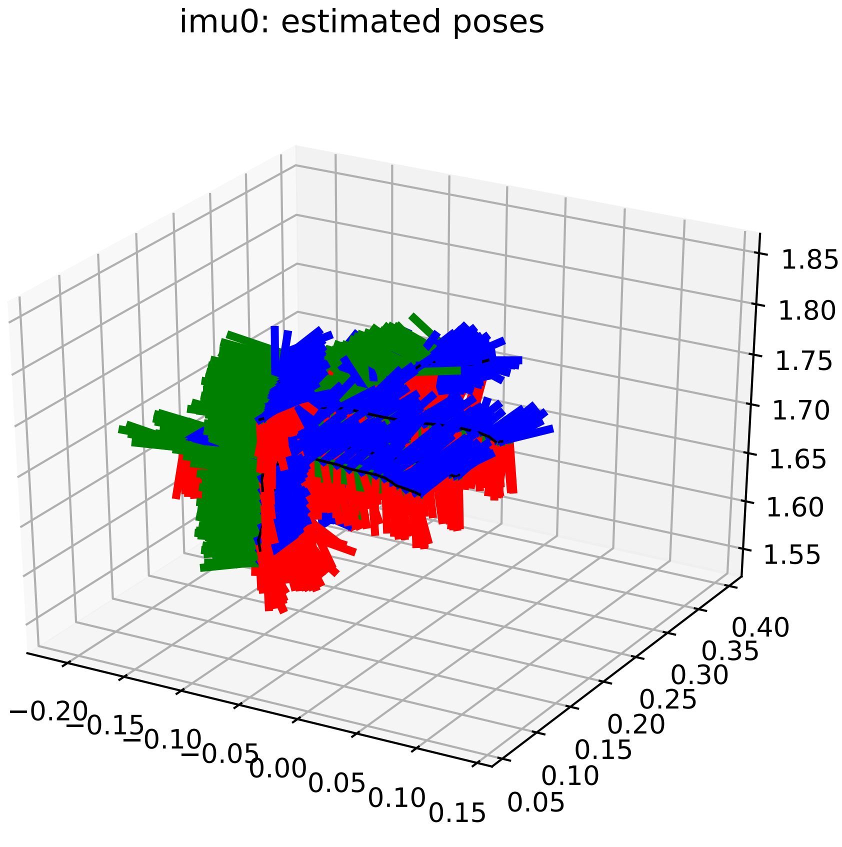
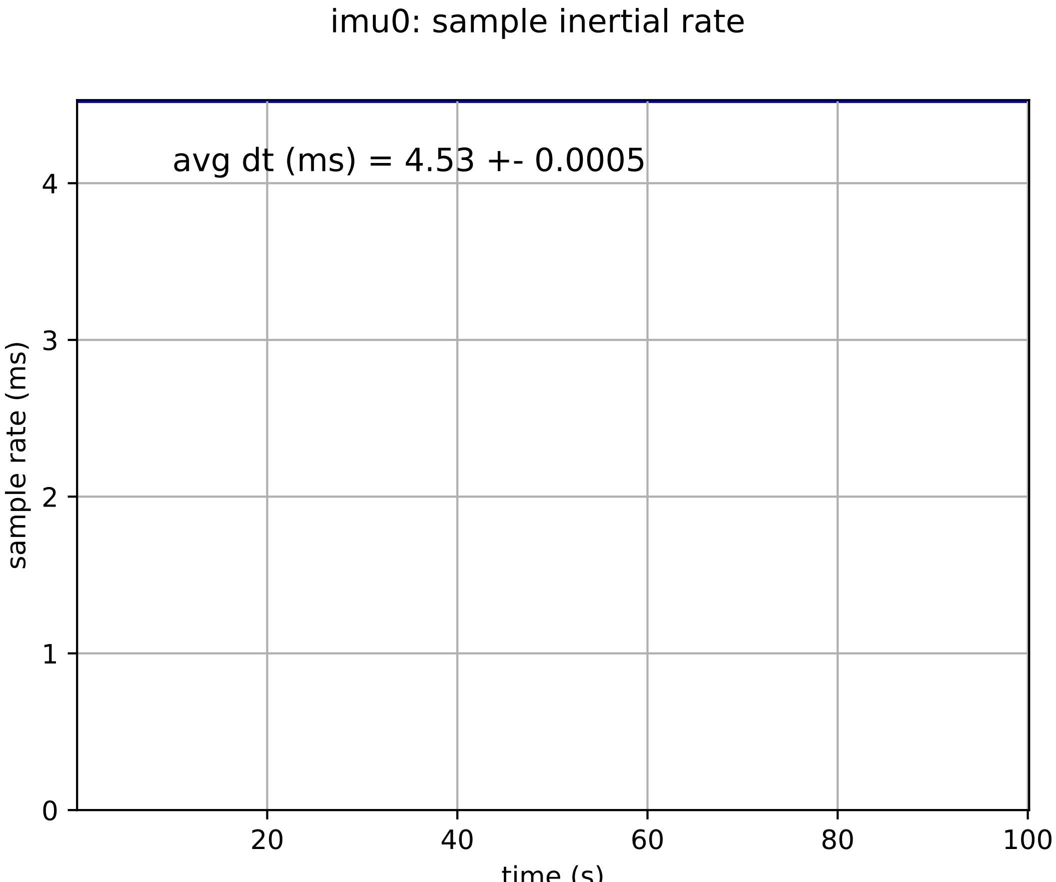
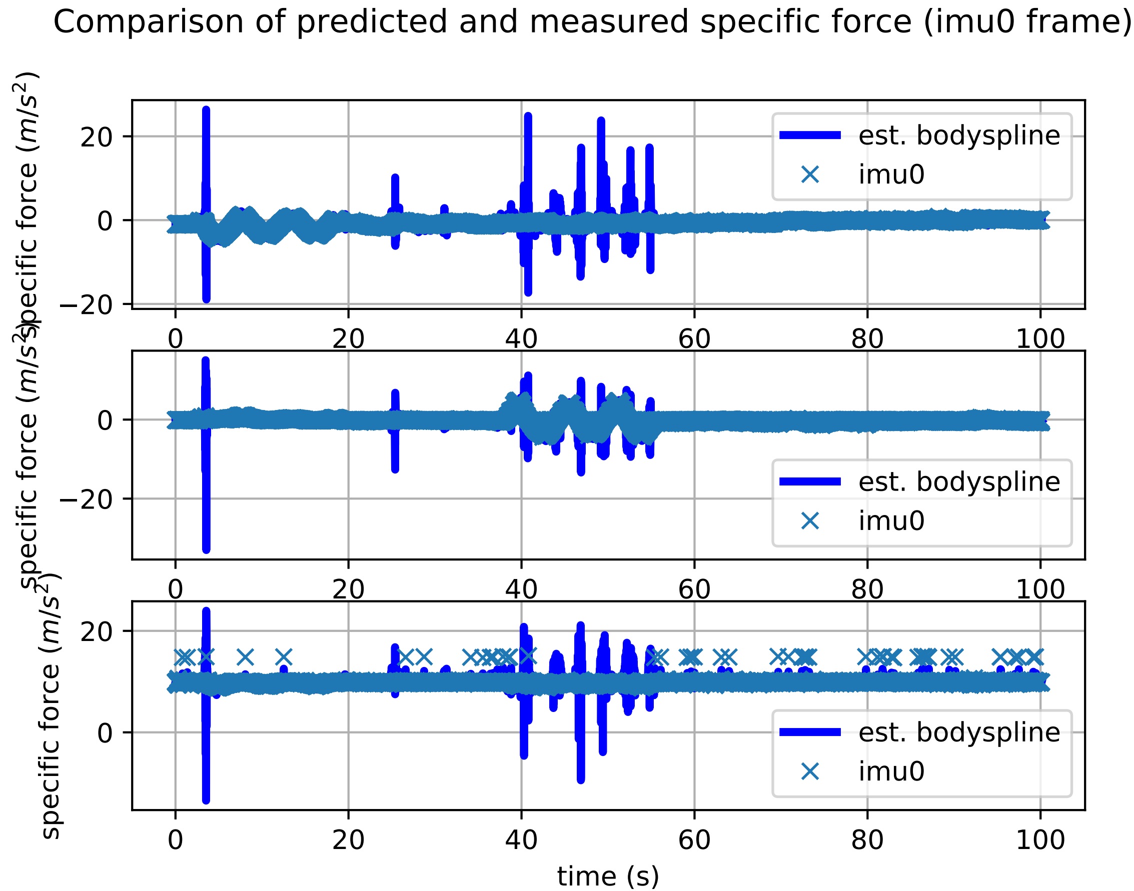
IMU0:

Model: calibrated
Update rate: 208.0

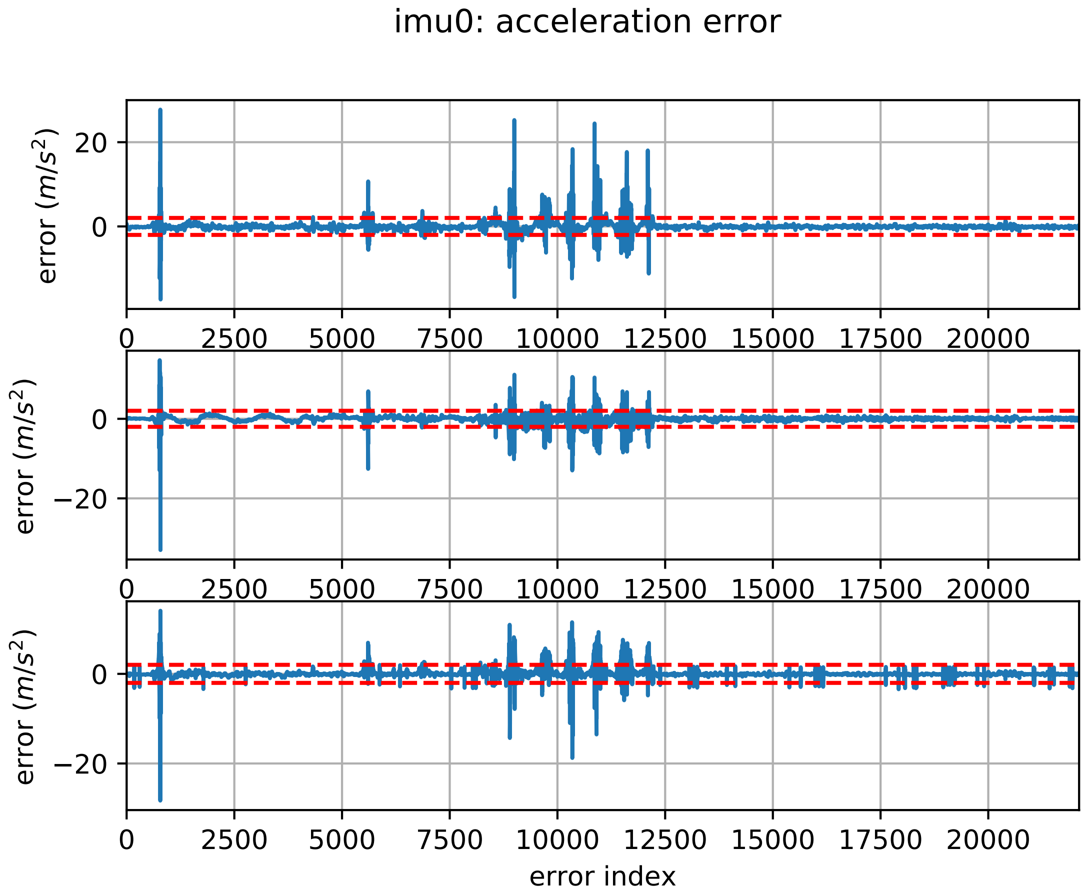
Accelerometer:
Noise density: 0.046615165871697874
Noise density (discrete): 0.6722934830586628
Random walk: 0.0013395309134963005
Gyroscope:
Noise density: 0.11900639970144332
Noise density (discrete): 1.716334704927665
Random walk: 0.0022961635833478244
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]


imu0: estimated poses
1.85
1.80
1.75
1.70
1.65
1.60
1.55
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15

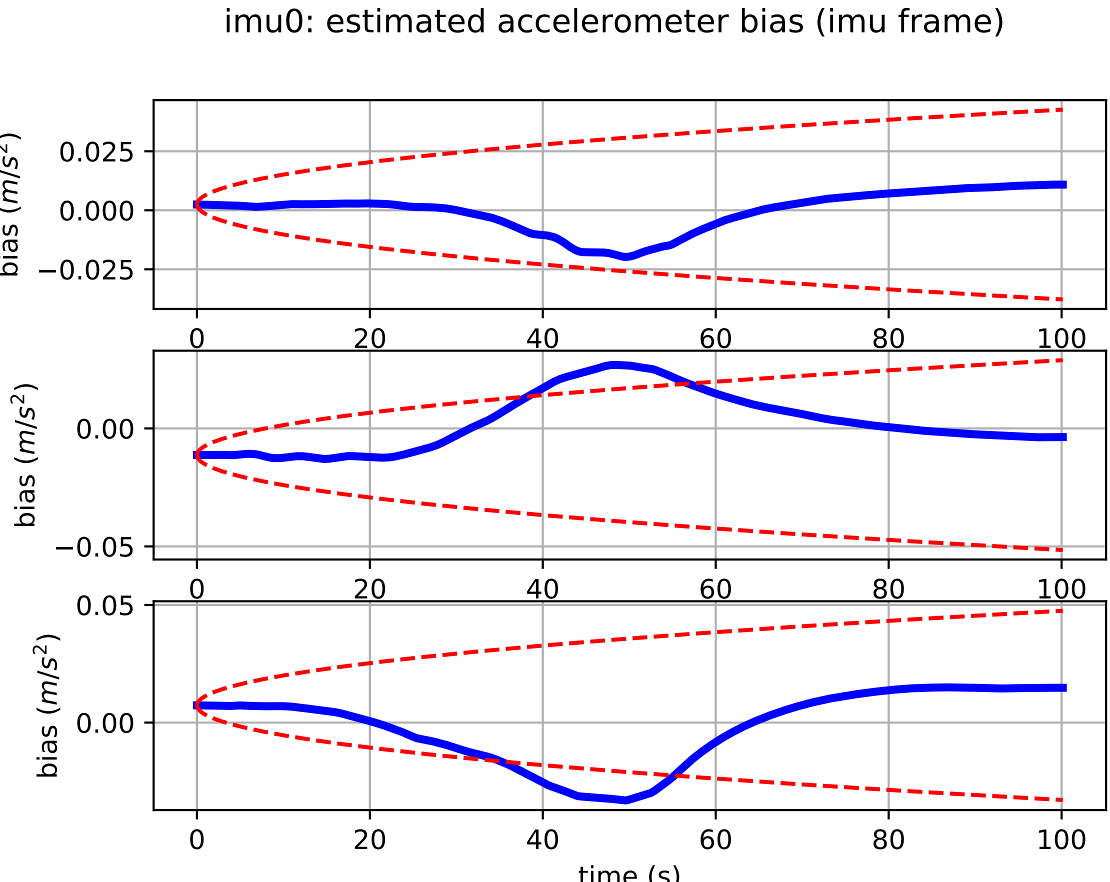

imu0: sample inertial rate
avg dt (ms) = 4.53 +- 0.0005
sample rate (ms)
time (s)
0
1
2
3
4
20
40
60
80
100



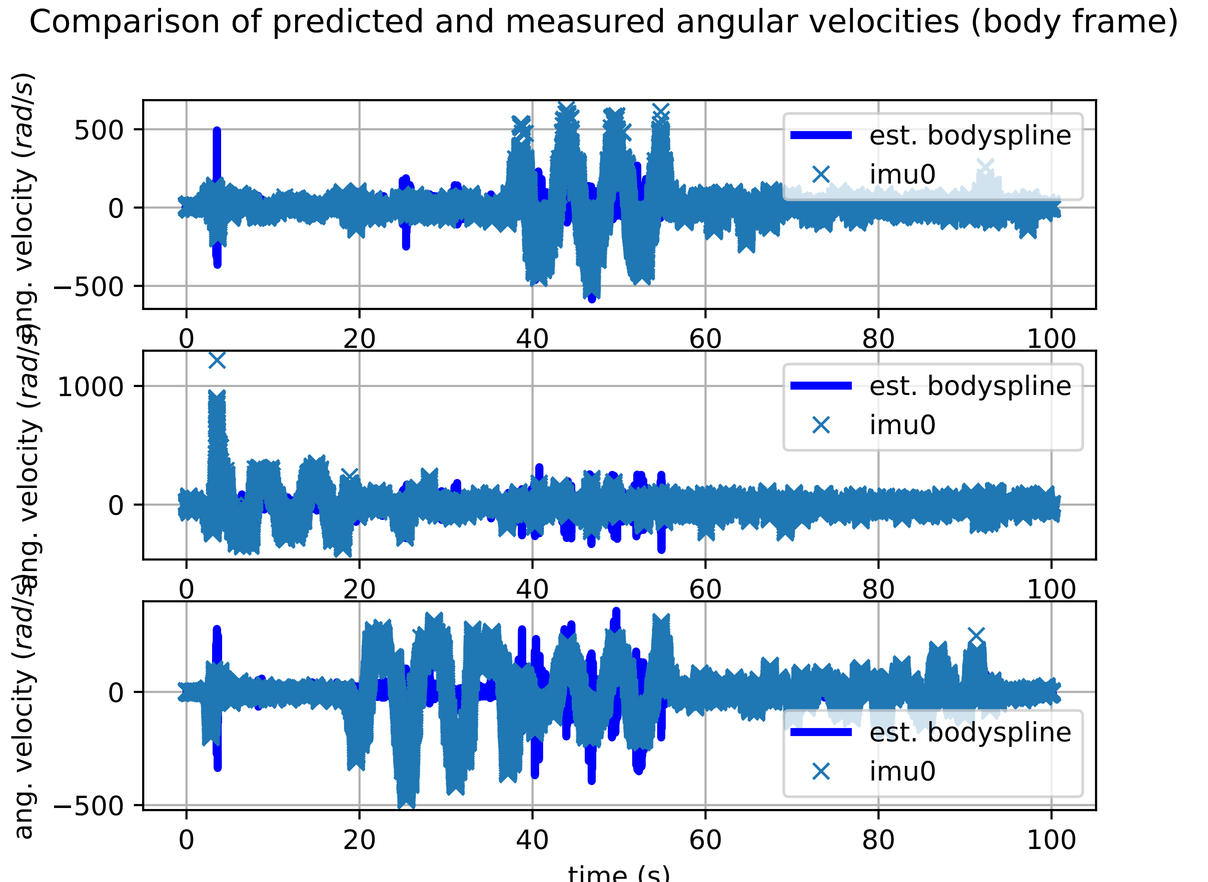
Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
20
0
−20
0
20
40
60
80
100
est. bodyspline
imu0
specific force (m/s^2)
0
−20
0
20
40
60
80
100
est. bodyspline
imu0
specific force (m/s^2)
20
0
0
20
40
60
80
100
est. bodyspline
imu0
time (s)



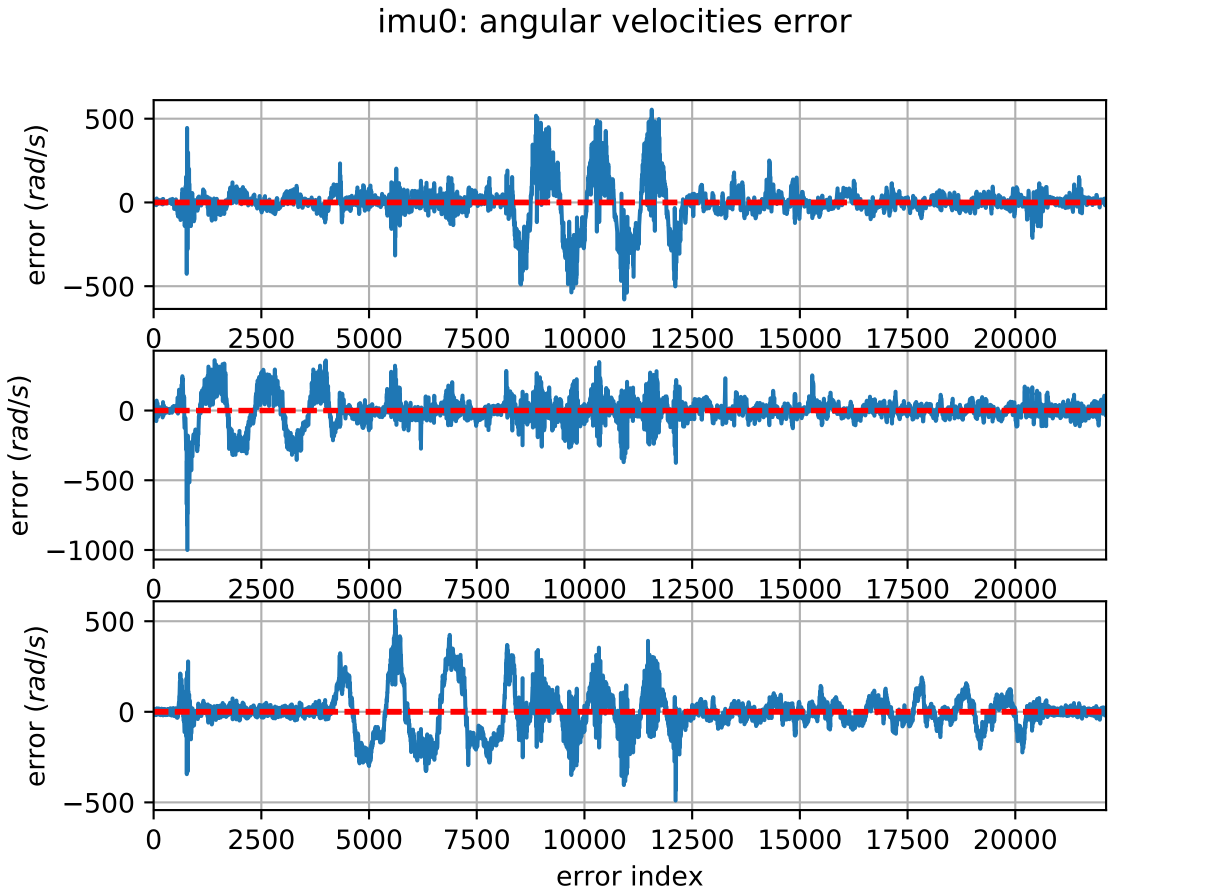
imu0: acceleration error
error (m/s^2)
20
0
0
2500
5000
7500
10000
12500
15000
17500
20000
error (m/s^2)
0
−20
0
2500
5000
7500
10000
12500
15000
17500
20000
error (m/s^2)
0
−20
0
2500
5000
7500
10000
12500
15000
17500
20000
error index

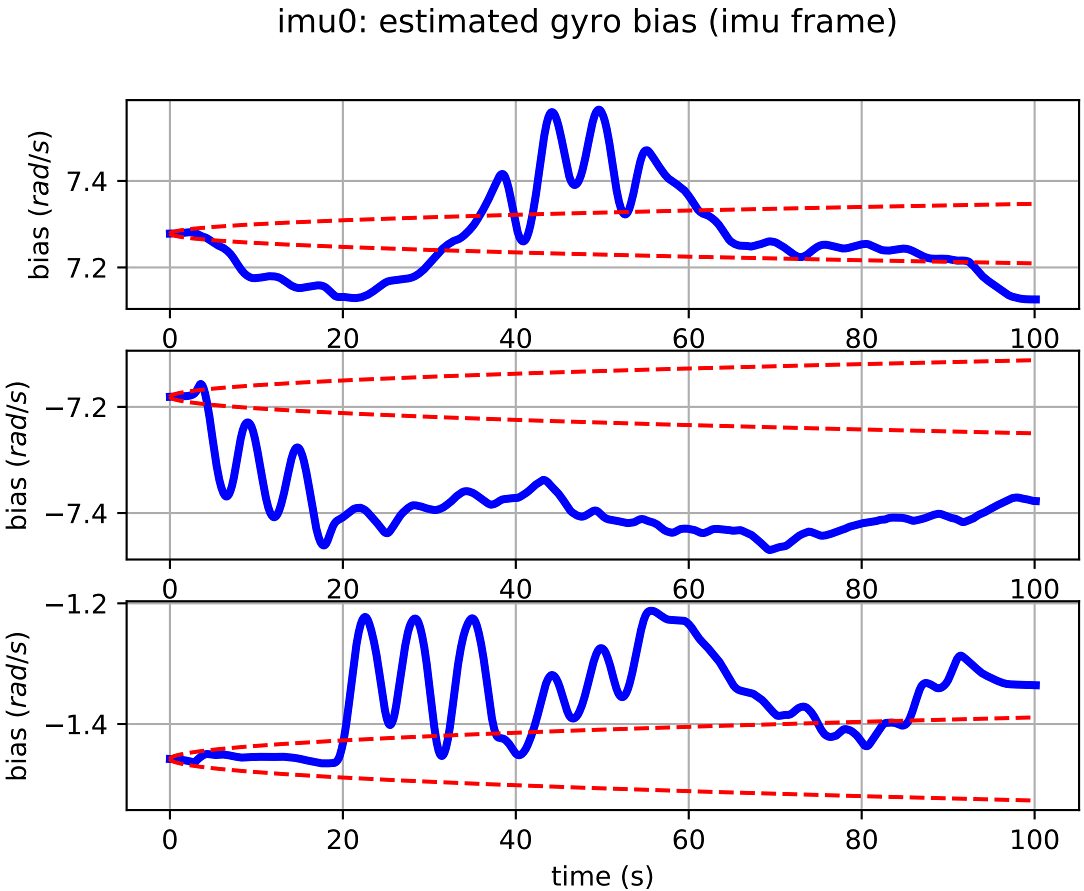


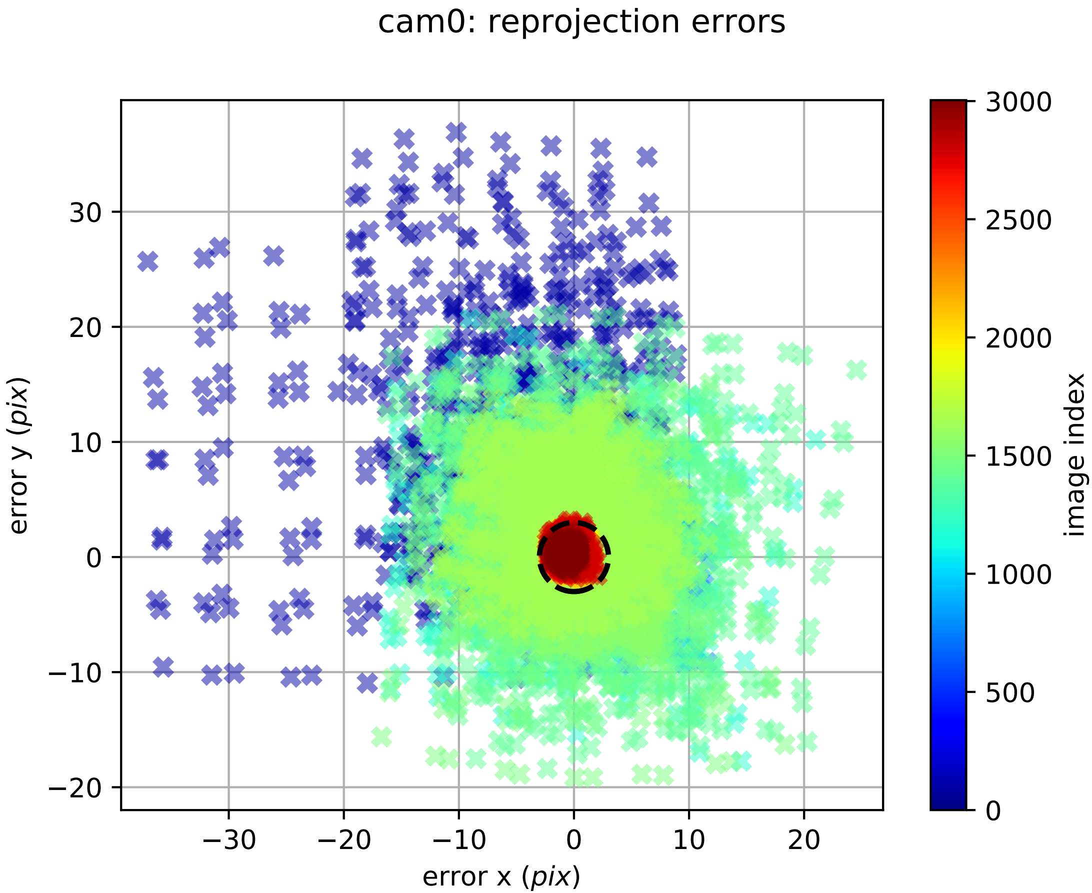
imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.025
0.000
−0.025
0.00
−0.05
0.05
0
20
40
60
80
100
time (s)

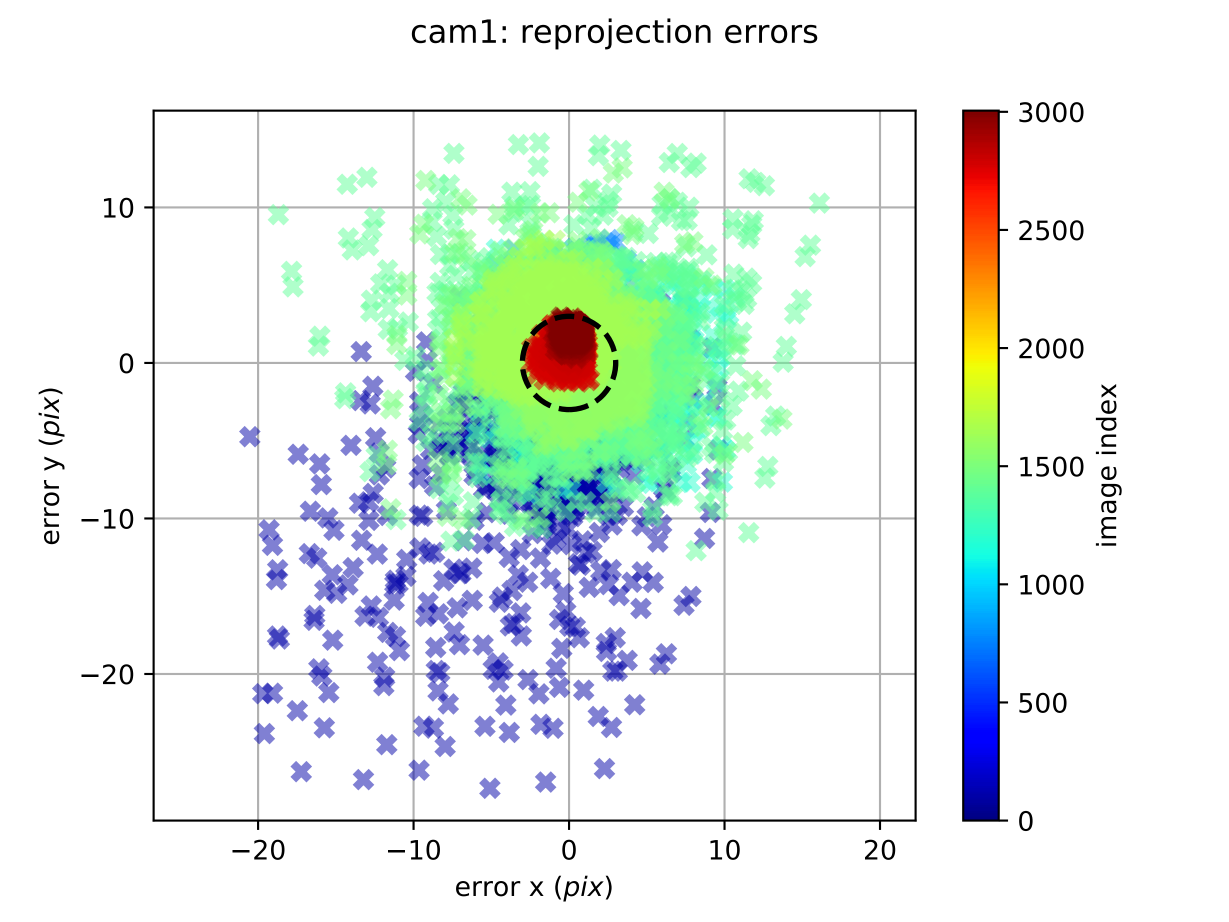


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
500
0
−500
1000
0
0
−500
0
20
40
60
80
100
time (s)
est. bodyspline
imu0


imu0: angular velocities error
error (rad/s)
error (rad/s)
error (rad/s)
500
0
−500
−1000
0
2500
5000
7500
10000
12500
15000
17500
20000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
7.4
7.2
0
20
40
60
80
100
bias (rad/s)
−7.2
−7.4
0
20
40
60
80
100
bias (rad/s)
−1.2
−1.4
0
20
40
60
80
100
time (s)


cam0: reprojection errors
error y (pix)
error x (pix)
image index
30
20
10
0
−10
−20
−30
−20
−10
0
10
20
3000
2500
2000
1500
1000
500
0


cam1: reprojection errors
error y (pix)
error x (pix)
10
0
−10
−20
−20
−10
0
10
20
image index
3000
2500
2000
1500
1000
500
0